Question: How many tensors may be created by repeated contraction of the tensor $T = T^{ij}_{kl}$?

Solution: Single contraction produces the following four mixed tensors:

1) for $i=k=v$: T^{vj}_{vl}

2) for $j=k=v$: T^{iv}_{vl}

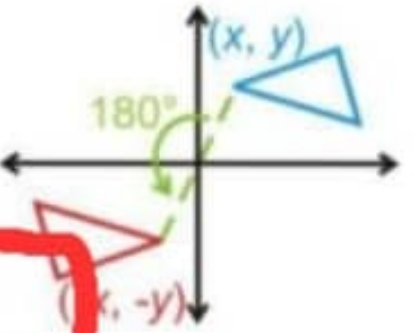
3) for $i=l=v$: T^{vj}_{kv}

4) for $j=l=v$: T^{iv}_{kv}

Question: Show that any contraction of the tensor $T = T^{i}_{jk}$ results in a covariant vector.

Solution: Consider the transformation

MCQs



Rotation of $180°$: $(x, y) \to (-x, -y)$

for $\theta = 180°$, rotation matrix becomes:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x+0 \\ 0-y \end{bmatrix} = \begin{bmatrix} -x \\ -y \end{bmatrix}$$

Implies under $180°$ rotation a point

$$(x, y) \longrightarrow (-x, -y)$$

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Rank/order of a tensor

The rank of a tensor is the number of indice
select each element of the tensor. Rank is a
In simple words, It is the total number of fr

For example,

1) (scalar) tensor of rank zero

2) v_α (vector) tensor of rank 1,

v^β (vector) tensor of rank 1

3) $T_{\mu\delta}$ (tensor) tensor of rank 2

4)

$A^{\mu}_{\nu\mu}$ (tensor) tensor of rank 1

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Long Q

$$\Rightarrow \mathcal{L} = \int_1^2 \frac{\sqrt{2}\,t}{\sqrt{t^2-1}}\,dt = \frac{\sqrt{2}}{2}\int_1^2 \frac{2t}{\sqrt{t^2-1}}\,dt$$

$$= \frac{2}{\sqrt{2}}\left|(t^2-1)^{1/2}\right|_1^2 = \sqrt{2}\,(\sqrt{3}) = \sqrt{6}$$

Question: ... the curve defined by $x^1 = 1$, $x^2 = t$; $1 \le t \le 2$ if the metric is that of the hyperbolic plane ($x^2 > 0$):

$$g_{11} = g_{22} = \frac{1}{(x^2)^2} \; ; \; g_{12} = g_{21} = 0$$

Solution:

consider the arc length formula:

$$\mathcal{L} = \int_1^2 \sqrt{g_{ij}\frac{dx^i}{dt}\frac{dx^j}{dt}}\,dt$$

$$\Rightarrow \mathcal{L} = \int_1^2 \left[g_{11}\left(\frac{dx^1}{dt}\right)^2 + g_{22}\left(\frac{dx^2}{dt}\right)^2\right]^{1/2} dt$$

$$= \int_1^2 \left[\frac{1}{t^2}(0) + \frac{1}{t^2}(1)^2\right]^{1/2} dt$$

$$= \int_1^2 t^{-1}\,dt = \ln t\Big|_1^2 = \ln 2 - \ln 1^{\nearrow 0}$$

$$\mathcal{L} = \ln 2$$

Question: Using Euclidean metric for cylindrical coordinates, calculate the length of arc along the circular helix $\bar{x}^1 = a\cos t$, $\bar{x}^2 = a\sin t$, $\bar{x}^3 = bt$ with a and b positive constants, from $t = 0$ to $t = c > 0$.

Solution: $\bar{x}^i \rightarrow$ for rectangular coordinates
$x^i \rightarrow$ for cylindrical coordinates

also

$\bar{x}^1 = x^1\cos x^2$, $\bar{x}^2 = x^1\sin x^2$, $\bar{x}^3 = x^3$

$$[g_{ij}] = \begin{bmatrix} 1 & 0 & 0 \\ & (x^1)^2 & \\ & & \end{bmatrix}$$

Given curve has parametric Eqs:

$\bar{x}^1 = a\cos t$, $\bar{x}^2 = a\sin t$, $\bar{x}^3 = bt$

$\Rightarrow$ $x^1 = a$, $x^2 = t$ and $x^3 = bt$ are the parametric Eqs of curve in cylindrical coordinates.

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Concept of Length and Ar

The norm (or length) of a

arbitrary vector V^2 or V_i

non - negative real numbe

$$\|\vec{V}\| = d\sqrt{V^2} = d\sqrt{V_i V^i}$$

or

$$ds^2 = g_{ij} dx^i dx^j$$

mcqs

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The inverse of the fundamental tensor

metric tensor

$$[g^{ij}]_{nn} = [g_{ij}]_{nn}^{-1}$$

is called the conjugate metric tensor.

$\star$ $g_{ij} = \hat{e}_i \cdot \hat{e}_j$ where $\hat{e}_i$ and $\hat{e}'_i$ are the

MCQs

Levi-Civita Symbol

Example: The permutation symbol satisfies :

i) $\varepsilon_{ijk}\varepsilon_{ijk} = 6$

ii) $\varepsilon_{ipq}\varepsilon_{jpq} = 2\delta_{ij}$

iii) $\varepsilon_{ijk}\varepsilon_{ijk} = 6$

where δ_{ij} is the Kronecker delta.

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$x^1 = t$, $x^2 = \cos t$, $x^3 = t^2 - 1$

Find the length of the arc for $1 \le t \le 2$.

Solution:

$$\mathcal{L} = \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}\, dt$$

where g_{ij} for spherical coordinates is:

Laplacian:

The Laplacian operator, also called th
harmonic operator, acting on a differentia
scalar function $\phi(x_i)$ is given by:

$$\Delta\phi = \nabla^2\phi = \delta_{ij}\frac{\partial\phi}{\partial x_i \partial x_j} = \frac{\partial^2\phi}{\partial x_i \partial x_i} = \nabla_{ii}\phi$$

$$= \partial_i \partial_i \phi$$

$$= \phi_{,ii}$$

MCQs

2) $\nabla \times \vec{r} = \vec{0}$

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and $\left|\frac{\partial \vec{r}}{\partial u_3}\right| = |\hat{i}| = 1$

Now we Sh... that the sy... consider

$\hat{e}_1 \cdot \hat{e}_2 = (-\sin u_1 \hat{j} + \cos u_1 \hat{k}) \cdot (co...$

$= \cos u_1 \sin u_1 + \cos u_1 \sin u_1$

$\Rightarrow \hat{e}_1 \cdot \hat{e}_2 = 0$

$\hat{e}_2 \cdot \hat{e}_3 = (\cos u_1 \hat{j} + \sin u_1 \hat{k})$

$\Rightarrow \hat{e}_2 \cdot \hat{e}_3 = 0$

and $\hat{e}_3 \cdot \hat{e}_1 = \hat{i} \cdot (-\sin u_1 \hat{j} +$

$\hat{e}_3 \cdot \hat{e}_1 = 0$

MCQs

Example: Check that the following system of elliptical cylindr... orthogonal or not?

$\vec{r} = a\cos\theta\,\hat{i} + b\sin\theta\,\hat{j} + z\hat{k} \; ; \; a \neq b$

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Nabla operator

The differential operator nabla often appear
the space of three variables it is defined as

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z}\hat{k}$$

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- Gradient → $\nabla\psi$
- Divergence → $\nabla . \vec{A}$
- Curl → $\nabla \times \vec{A}$
- Laplacian → $\nabla^2 \psi$

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